

1 Experimental Setup

The experimental setup consists of a single-phase autotransformer, the transformer under test, voltage and current measurement instrumentation, and a digital storage oscilloscope.

The autotransformer is used to control the excitation voltage applied to the transformer winding. The resulting voltage and magnetizing current waveforms are captured using the DSO and exported for subsequent analysis.

The experimental investigation is performed on three transformer cores with different ratings and core constructions.

2 Transformer Specifications

The transformers investigated during the experiment are listed in Table 1.

Property	Transformer 1	Transformer 2	Transformer 3
Input voltage	115 V AC	230 V AC	230 V AC
Output voltage	230 V AC	12–0–12 V AC	15–0–15 V AC
Power rating	1.5 kVA	48 VA	30 VA
Primary turns	99	400	1500
Secondary turns	196	45	200
Core material	CRNO Silicon Steel	CRGO Silicon Steel	CRGO Silicon Steel
Core area	5000 mm ²	2023 mm ²	496 mm ²
Mean magnetic length	435 mm	220 mm	130 mm

Table 1: Specifications of the transformers investigated.

3 Theoretical Background

When an alternating voltage is applied to a transformer winding, an alternating magnetic flux is established in the core. The relationship between the applied winding voltage and magnetic flux is given by

$$v(t) = N \frac{d\Phi(t)}{dt} \quad (3.1)$$

where $v(t)$ is the winding voltage, N is the number of winding turns, and $\Phi(t)$ is the magnetic flux.

The magnetic flux density is related to the flux by

$$B(t) = \frac{\Phi(t)}{A_c} \quad (3.2)$$

where A_c is the effective cross-sectional area of the core.

The magnetomotive force produced by the winding current is

$$\mathcal{F} = Ni \quad (3.3)$$

and the corresponding magnetic field intensity is

$$H = \frac{Ni}{l_m} \quad (3.4)$$

where l_m is the mean magnetic path length.

Therefore, the experimentally measured winding voltage and current can be used to determine the corresponding B and H values.

Now From,

$$v(t) = N \frac{d\Phi}{dt}, \quad (3.5)$$

the magnetic flux can be obtained by integration:

$$\Phi(t) = \frac{1}{N} \int v(t) dt. \quad (3.6)$$

Consequently,

$$B(t) = \frac{1}{NA_c} \int v(t) dt. \quad (3.7)$$

This relationship is used to obtain the magnetic flux density from the measured voltage waveform.

The magnetic field intensity is obtained from the measured winding current:

$$H(t) = \frac{Ni(t)}{l_m}. \quad (3.8)$$

Thus, the measured current waveform provides the horizontal-axis quantity required for reconstructing the B – H characteristic.

The area enclosed by the B – H loop represents the hysteresis energy loss per unit volume per cycle:

$$w_h = \oint H dB. \quad (3.9)$$

For a core of volume V_c operating at frequency f , the corresponding hysteresis power loss is

$$P_h = fV_c \oint H dB. \quad (3.10)$$

This expression provides the basis for estimating the hysteresis loss from the experimentally reconstructed B – H loop.

4 Experimental Methodology

1. Maintain the transform at no load condition and excite the primary with the rated voltage (Figure 1)
2. Acquire the steady-state voltage and current waveforms using the digital storage oscilloscope as csv data.
3. Use the attached Matlab script to post process the acquired data.

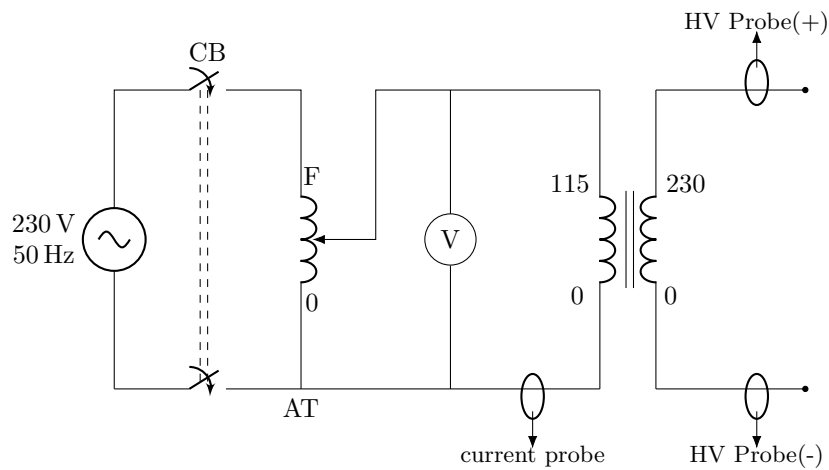


Figure 1: Experimental setup for voltage–current characterization and hysteresis-loop tracing of Transformer .

5 Results

The B–H curve tracing was done for three different transformers available in the lab:

1. 1.5 kVA, 115 V to 230 V Transformer
2. 48 VA, 230 V to 12–0–12 V Centre-Tapped Transformer
3. 30 VA, 230 V to 15–0–15 V Centre-Tapped Transformer

The calculated hysteresis power losses for these transformers are shown in Table 2.

Sl. No.	Transformer Specification	Hysteresis Loss (W)
1	1.5 kVA, 115 V / 230 V	40.1
2	48 VA, 230 V / 12–0–12 V Centre-Tapped	2.4
3	30 VA, 230 V / 15–0–15 V Centre-Tapped	1.6

Table 2: Summary of hysteresis losses in different transformers

Experimental results for 1.5 KVA Transformer

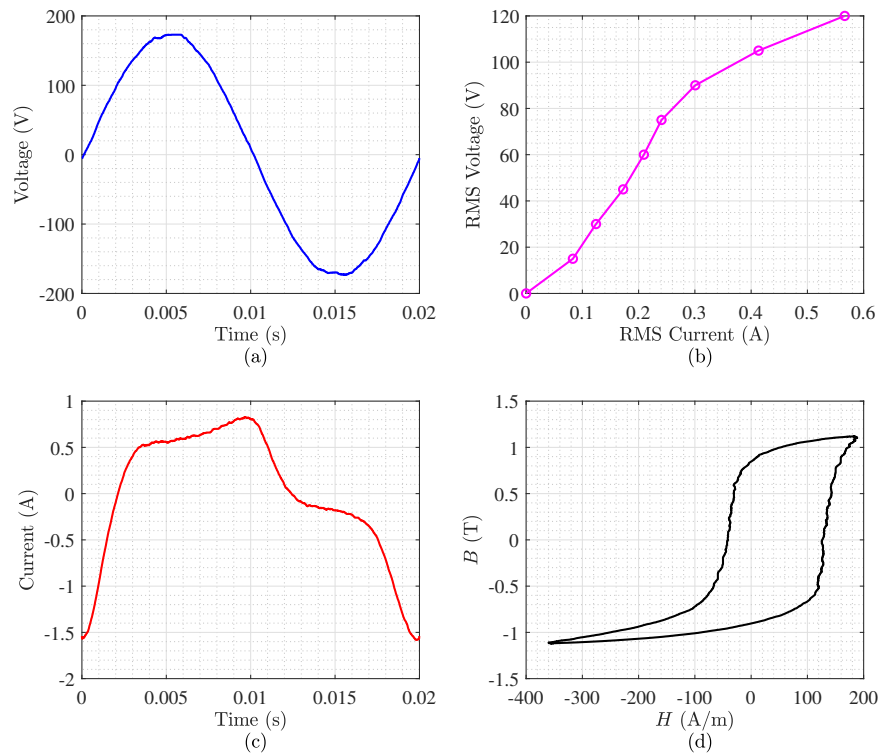


Figure 2: (a) Input primary voltage (b) VI characteristics (c) Exciting current (d) BH curve

Experimental results 48 VA & 30 VA Transformer

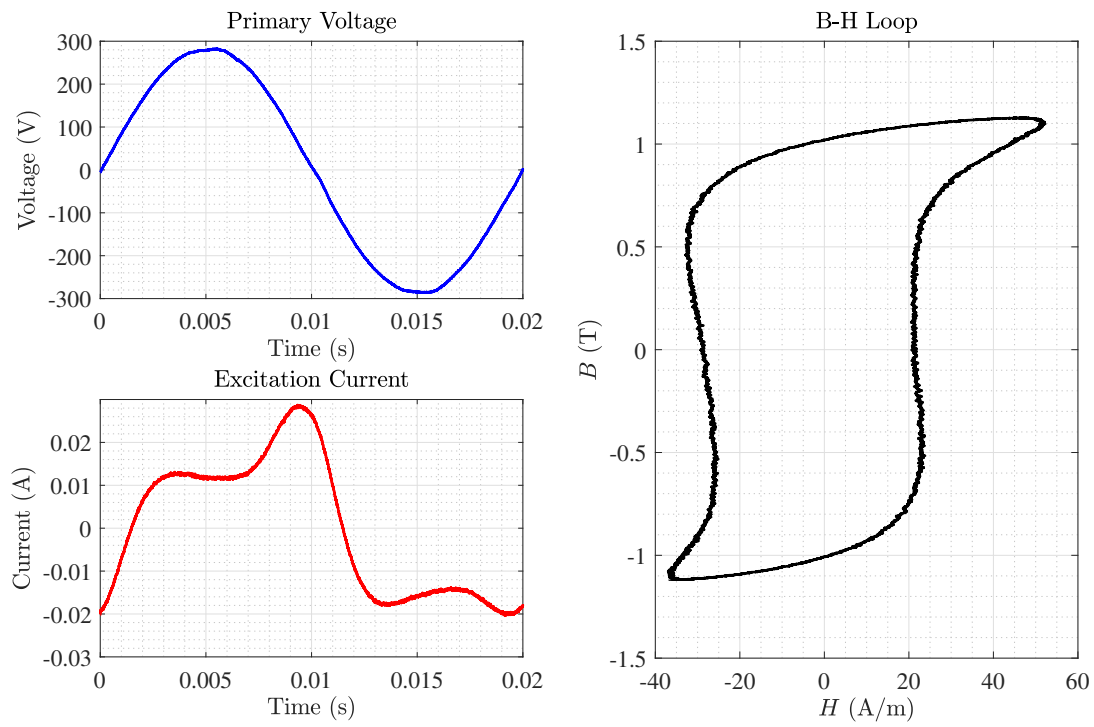


Figure 3: 48 VA Transformer : (a) Input primary voltage (b) VI characteristics BH curve

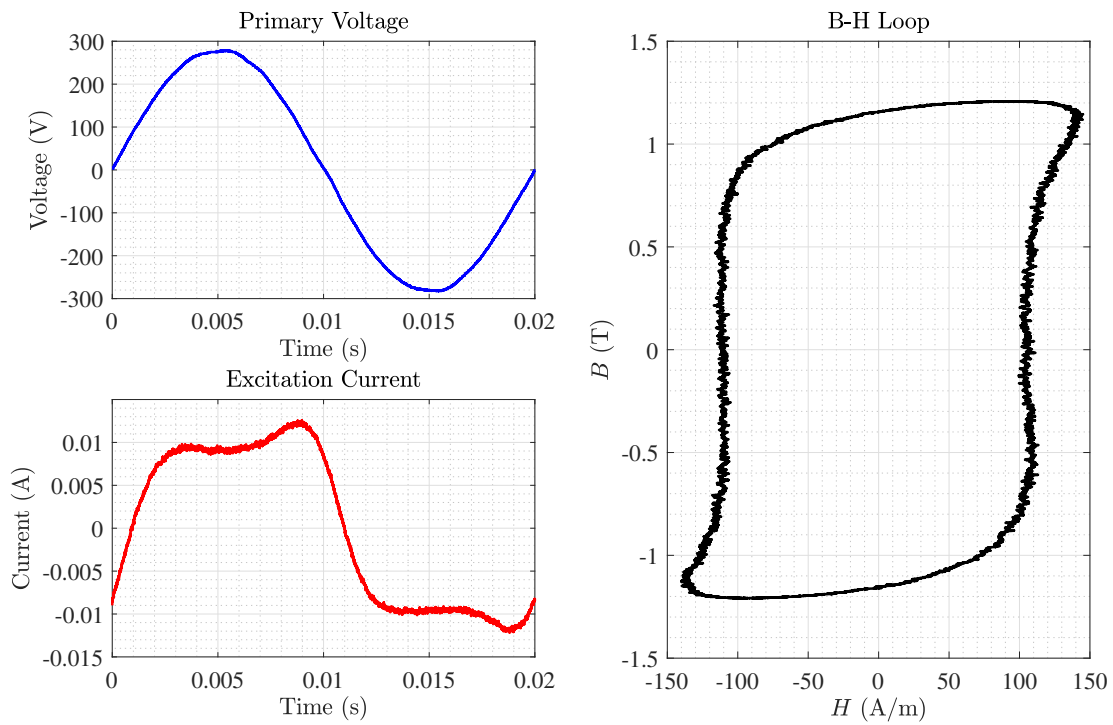


Figure 4: 30 VA Transformer :(a) Input primary voltage (b) VI characteristics (c) BH curve

6 MATLAB Code for B-H Hysteresis Characterization

The following MATLAB script was used for processing the oscilloscope data and obtaining the magnetic hysteresis loop and hysteresis loss of the transformer core.

```

1 %% EE2020A
2 % Code to trace the hysteresis chara of a given magnetic system
3 % system : Procon Transformer
4 % Author : Harikrishnan S, IIT Palakkad
5 % Date : 29/08/2024 (Updated 07/08/2025)
6
7 %% Note :
8 % 1) This code is a generic code, students are encouraged to go
   through this
9 % code line by line and make necessary corrections where ever
   necessary.
10
11 % 2) Make sure you save this file and the csv files in a single
   directory
12
13 % 3) For this code channel 1 data is taken as voltage information
   while
14 % channel 2 data is taken as current information
15
16 clc;
17 clear;
18 close all;
19
20 % ----- Transformer & Core Parameters
   -----%
21 N1 = 99; % Primary turns
22 N2 = 196; % Secondary turns
23 freq = 50; % Supply frequency in Hz
24 w0 = 2 * pi * freq;
25
26 Ae = 5000e-6; % Core cross-sectional area (m^2)
27 lm = 0.435; % Magnetic path length (m)
28 volume = Ae * lm; % Core volume (m^3)
29
30 % ----- Read Oscilloscope Data -----%
31 filename = 'SDS00033.csv'; % Replace with your actual file
32 data = readtable(filename);
33
34 time = data.Source; % Time vector from scope
35 voltage = data.CH1; % Voltage waveform (CH1 - secondary)
36 current = data.CH2; % Current waveform (CH2 - excitation

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```

    current)
37
38 sample_size = abs(time(3) - time(2));
39 n           = length(time);
40
41 % ----- Crop One Cycle for Analysis -----%
42 limit1 = 1515;                                % choose your start value
43 limit2 = round(limit1 + ((20e-3)/sample_size)-1);
44
45 v_sec   = voltageex(limit1:limit2);            % CH1: Secondary voltage
46 i_exc   = currentx(limit1:limit2);            % CH2: Exciting current
47
48 t       = 0:sample_size:(length(v_sec)-1)*sample_size;
49 v_prim  = (N1 / N2) * v_sec;                   % Reconstructed primary
    voltage
50
51 % ----- Smoothing & Detrending -----%
52 v_prim  = smooth(v_prim, 15);
53 i_exc   = smooth(i_exc, 15);
54
55 v_prim  = v_prim - mean(v_prim);
56 i_exc   = i_exc - mean(i_exc);
57
58 % ----- Flux Linkage by Numerical Integration -----%
59 lambda  = cumtrapz(t, v_prim);                 % Units Wb
60 lambda  = lambda - mean(lambda);
61
62 % ----- Compute B and H -----%
63 B = lambda / (N1 * Ae);                        % Magnetic flux density (
    T)
64 H = (N1 * i_exc) / lm;                         % Magnetic field strength
    (A/m)
65
66 % ----- Compute Losses -----%
67 W_hyst_per_vol = trapz(B, H);                 % Area under B-H per
    cycle
68 P_hyst = W_hyst_per_vol * freq * volume;      % Hysteresis loss in
    watts
69
70 % ----- Plotting -----%
71 figure('Color', 'w', 'Position', [100 100 1000 600]);
72
73 % Define styling
74 fontSize = 12;

```

```

75 lineWidth = 1.5;
76
77 % Top-left: Primary Voltage
78 subplot(2,2,1);
79 plot(t, v_prim, 'b', 'LineWidth', lineWidth);
80 xlabel('Time (s)', 'Interpreter', 'latex', 'FontSize', fontSize);
81 ylabel('Voltage (V)', 'Interpreter', 'latex', 'FontSize', fontSize);
82 title('Primary Voltage', 'Interpreter', 'latex', 'FontSize',
      fontSize);
83 grid on; grid minor;
84 box on;
85
86 % Bottom-left: Excitation Current
87 subplot(2,2,3);
88 plot(t, i_exc, 'r', 'LineWidth', lineWidth);
89 xlabel('Time (s)', 'Interpreter', 'latex', 'FontSize', fontSize);
90 ylabel('Current (A)', 'Interpreter', 'latex', 'FontSize', fontSize);
91 title('Excitation Current', 'Interpreter', 'latex', 'FontSize',
      fontSize);
92 grid on; grid minor;
93 box on;
94
95 % Right side (both rows): B-H Loop
96 subplot(2,2,[2 4]);
97 plot(H, B, 'k', 'LineWidth', lineWidth);
98 xlabel('$H$ (A/m)', 'Interpreter', 'latex', 'FontSize', fontSize);
99 ylabel('$B$ (T)', 'Interpreter', 'latex', 'FontSize', fontSize);
100 title('B-H Loop', 'Interpreter', 'latex', 'FontSize', fontSize);
101 grid on; grid minor;
102 box on;
103
104 % ----- Display Results -----%
105 fprintf('=====\n');
106 fprintf('Hysteresis Loss Power : %.4f W\n', P_hyst);
107 fprintf('=====\n');

```